

Faris Babsail

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EDUCATION

University of California, Berkeley

Bachelor of Science in Mechanical Engineering - GPA: 3.8/4.0

Berkeley, CA

Aug. 2023 – May 2027

EXPERIENCE

Consultant Intern

Oliver Wyman

Jun. 2026 – Sep. 2026

Riyadh, Saudi Arabia

- Analyzed 80K+ delinquent establishments to design 1K- and 3K-account collection cohorts for a leading Middle Eastern social insurer, helping recover \$7.5M of \$29.1M outstanding (26%)
- Rebuilt the client's legal-case reporting from ad hoc manual files into an automated dashboard covering \$1.5B+, defining a source and owner for every field
- Led thrice-weekly cross-functional pilot meetings to track execution, resolve issues, and align next steps across client departments

Project Analyst Intern

Saudi Industrial Development Fund (SIDF)

Jun. 2025 – Aug. 2025

Riyadh, Saudi Arabia

- Evaluated a major automation upgrade at a publicly listed manufacturer, delivering NPV and payback scenarios to decision-makers
- Modeled automation ROI and built CapEx and OpEx benchmarks to support industrial project appraisals
- Built a per-unit utility-cost tool across cement, steel, and paper, accelerating utility checks during credit reviews

Research Assistant

King Abdullah University of Science and Technology (KAUST)

Jul. 2022 – Jan. 2023

Thuwal, Saudi Arabia

- Increased hydrogen internal combustion engine efficiency by 20% using PCC technology under Prof. Im
- Used CHEMKIN to analyze laminar flame speed and ignition delay for ultra-lean hydrogen combustion
- Presented combustion findings at KAUST and the National Olympiad for Scientific Creativity

LEADERSHIP & PROJECTS

Combat Robotics Club Team Member

Combat Robotics Club, University of California Berkeley

Aug. 2024 – May 2026

Berkeley, CA

- Placed second in the 1 lb division after designing and iterating a custom robot using CAD and FEA to improve durability and performance
- Manufactured and assembled robot components using machining, 3D printing, and soldering; piloted the robot during competition

Exoskeleton Team Member

Berkeley Nanotech Lab, University of California Berkeley

Aug. 2024 – May 2026

Berkeley, CA

- Designed and built a wearable exoskeleton in CAD, balancing functionality, fit, and user comfort
- Tested the exoskeleton on users and used performance data to guide design improvements

MusicPod: Tripod-Mounted Music / Media Stand

Jan. 2026 – May 2026

- Designed and fabricated a folding tripod-mounted stand using FDM-printed PLA components and 6061-T6 aluminum support shafts; developed GD&T drawings and fit/tolerance specifications
- Iterated the design after prototype failures by replacing 3D-printed pins with aluminum shafts, redesigning the hinge mechanism, and adding a universal 1/4-20 tripod interface

Sound-Tracking Robot

Jan. 2025 – May 2025

- Built an ESP32 mobile robot with hard-real-time multitasking in ESP-IDF for the ME135 Capstone
- Designed and 3D-printed a lightweight chassis with a servo-steered mic array for 360° sound tracking
- Coded a desktop GUI for live tracking data, commands, and system debugging

SKILLS

Mechanical & Manufacturing: SolidWorks/CAD, FEA, GD&T, Tolerancing & Fits, Material/Process Selection, Manual Mill, Lathe, Bandsaw, FDM/3D Printing, Soldering, Electronics Assembly

Robotics, Controls & Testing: ROS 2, Python, MATLAB, Simulink, LabVIEW, CHEMKIN, ESP32/ESP-IDF, Arduino, Robot Kinematics, PID Control, Instrumentation, Data Acquisition, Experimental Design, Signal Processing